

Dear Editors:

We would like to submit the enclosed manuscript entitled “Options on Youth: Generative Artificial Intelligence Uncertainty and the Contingent Turn in Entry-Level Hiring” for consideration in the Special Issue “Systems Approaches to Generative AI: Workforce Development, Organisational Learning, and Economic Transformation” of *Systems*. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

Almost the entire debate about generative artificial intelligence and young workers is about capability: will the models do what juniors do? This manuscript argues that what actually closes the entry port is not capability but its dispersion. Hiring a school leaver is a partly irreversible investment, because the firm sinks a training cost it cannot recover if the frontier moves against the task bundle it hired for. The theory of investment under uncertainty then says the firm should raise its hurdle and wait, so the entry-level labour market can contract before the technology has displaced a single task. Using 29,660 firm-year observations for 3,200 Chinese A-share listed firms over 2016–2025, we show that it does.

The highlights of the paper are as follows.

(1) Dispersion, not level. One standard deviation of firm-level generative-AI uncertainty lowers entry-level vacancy creation by 1.230 vacancies per thousand employees, 15.6% of the mean, while the level of adoption enters the same equation with the opposite sign (+0.300) and general economic policy uncertainty is 6.7 times smaller. Studies that measure only adoption average across two channels that work in opposite directions, which is one reason their findings disagree.

(2) A sharp discriminating test. The same coefficient for vacancies requiring five or more years of experience is -0.090 , 13.7 times smaller, and a cross-equation test on the stacked model rejects equality decisively. General pessimism, a demand shock or poor performance cannot produce that asymmetry; an option whose value scales with what the firm cannot recover can.

(3) The entry port is not closing, it is becoming revocable. Uncertainty raises the contingent share of entry-level vacancies — internships, agency placements, short fixed-term contracts — by 2.25 percentage points, and the shift is largest exactly where the reversible margin is cheapest to use. A study that counts young employees will therefore miss most of what is happening, because the first thing a firm gives up is not the young worker but the commitment to them.

(4) Irreversibility, inaction and hysteresis. The chill is 3.5 times larger for firms one standard deviation above the mean of firm-specific training intensity than one standard deviation below it, rises with employment protection, and is attenuated where reversible employment is available. Uncertainty also widens a zone of inaction: it raises the probability of no youth adjustment at all by 0.050 and lowers the probability that an already inactive firm resumes expansion by 0.126.

We believe the manuscript speaks directly to the aims of the Special Issue, and that its policy implication is unusually specific. If the entry port is closing because firms are uncertain rather than because they have been displaced, the cheapest available intervention is not retraining but information: sectoral technology roadmaps, disclosure standards for model capabilities and limitations, and certification of what models can be relied upon all narrow the dispersion without touching the frontier, and are therefore youth employment policy. Identification rests on two shift-share instruments and a peer-diffusion instrument that pass the Kleibergen–Paap and overidentification tests, on a dynamic difference-in-differences design with flat pre-shock leads, and on a wild cluster bootstrap, randomisation inference and coefficient-stability bounds; every number in every table is produced by the released estimation code. We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to the address below.

Thanks very much for your attention to our paper.

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Very sincerely yours,
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